



Experimental and modeling study for the effect of methanol blending on ammonia oxidation and NO_x formation at high pressure

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ABSTRACT

To investigate the kinetic of ammonia-methanol mixtures oxidation and NO_x formation under low-to-intermediate-temperature and high-pressure conditions, the mole fractions of key intermediates and products in NH₃/CH₃OH were measured in a flow reactor, at 5.0 MPa, 650–1250 K. A comprehensive kinetic model was proposed with the modification of C-N interaction mechanism. The present model can adequately predict the species concentration, laminar flame speed and ignition delay time of ammonia-methanol mixture. The study indicated that under current conditions NO and N₂O change non-monotonic with temperature. There is a trade-off between NO and N₂O at intermediate temperature range. NO increases as N₂O begins to decrease with temperature. Based on a kinetic analysis aided by the developed model, the HO₂ content is identified as the key factor influencing the overall reactivity at high pressure. H-elimination reactions of CH₃OH and CH₂O by NH₂ enhances the consumption of methanol and further promotes ammonia oxidation. However, with higher methanol content, this reaction transforms NH₂ to NH₃ at the same time and exhibits an inhibition effect. As to NO formation, H₂NO plays a crucial role, acting as chain carrier and converts NH₂ to NO. The competition of NO formation and reduction results in the non-monotonic trend of NO concentration, while the further decomposition of N₂O at high temperature leads to the decrease of N₂O concentration. The above trend can be enhanced by methanol addition, as the dominant reactions involved in the rate production of NO, such as NH₂ + NO₂ = H₂NO + NO, followed by HONO (+M) = NO + OH (+M) and HNO + O₂ = HONO + NO, are strengthened with more methanol addition.

1. Introduction

Ammonia (NH₃) is an attractive zero-carbon fuel and hydrogen carrier and it has received widespread attention [1]. Ammonia has been extensively used in chemistry industries and fertilizer, which enabled highly developed generation, storage and transport technologies. In addition, ammonia is an ideal hydrogen carrier with a high hydrogen storage density and easy to be transported. However, there are challenges encountered in the application of ammonia in thermal and power equipment, such as difficulties in ignition, slow combustion rates, poor flame stability and high NO_x emission [2,3]. Solutions to overcome the bottleneck associated with direct ammonia combustion are proposed that introducing the reactive fuels into the mixture seems one best solution with low cost, which is also controlled flexibly. Previous studies

have extensively addressed that Hydrogen [4,5], syngas, methane [6,7], dimethyl ether (DME) [8,9] and methanol, etc. are good choice.

Profoundly grasping the combustion behavior and kinetic mechanism of ammonia/additives combustion is the key to the success in combustor design. Therefore, it is necessary to develop a reliable chemical mechanism to accurately predict experimental data and elucidate the chemical kinetic perspective. Fundamental researches on laminar burning velocity (LBV), ignition delay time (IDT), and species formation (SF) in well-defined experimental condition are required for model validation and kinetic mechanism revealing. So far, there are sufficient data of LBV, IDT and SF under variable conditions of NH₃+H₂, NH₃+CO/syngas, and NH₃+CH₄, NH₃+DME.

However, renewable fuels such as methanol (CH₃OH) are not well studied, even though methanol is one of the most favorable substitutes

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for the diesel and gasoline in the engines. Methanol engines can operation stably under a range of temperature, equivalence ratios and power levels [10] with low exhaust emissions of NO_x and CO_2 . Research from Vancoillie et al. [11] indicated that thermal efficiency of methanol engine is up to 42 % with lower or comparable emission levels than gasoline. In addition, methanol can be produced from hydrogen and CO_2 to form a carbon neutral green fuel with mature product process. Furthermore, it has strong promoting effect on ammonia combustion. IDT measured by He et al. [12] indicated that the promotion on IDT of 5 % methanol addition to ammonia is much more effective than methane and slightly higher than hydrogen. Methanol addition promotes reactivity through the formation of CH_2OH and HCO radicals, and it also produces free radicals (OH and HO_2) that facilitate ammonia oxidation. Therefore, it is essential to study the chemical mechanism of ammonia-methanol mixture.

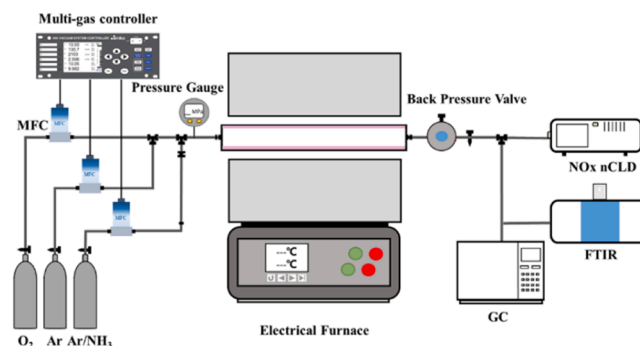
Wang et al. [13] measured the laminar flame speeds of ammonia mixed with methanol (0 %–100 % in volume) at different equivalence ratio (0.7–1.8) and 1 atm using heat flux method. They also developed a new chemical kinetic model, CEU- NH_3 , based on their previous NH_3 -syngas mechanism, which captures the measured data well. As to IDT, Li et al. [14] measured the ignition delay times of ammonia-methanol in a rapid compression machine (RCM) at elevated pressure from 20 to 40 atm with methanol mole fractions from 0 % to 100 %. A kinetic model was established and interaction reactions between two fuels were included. This model showed quite well capability in predicting the IDTs of mixtures when methanol ≥ 5 %, whereas it underestimated the IDT of ammonia mixed with 1 % methanol under stoichiometric and fuel-rich condition. Nadiri et al. [15] also measured the IDTs of ammonia mixed with 0–20 % methanol in content in RCM and a kinetic model was generated automatically by RMG, which predicted the measured LBV and IDT of ammonia-methanol mixtures satisfactorily. For SF, Zhuang et al. [16] investigated the oxidation kinetics of ammonia-methanol mixtures using the SVUV photoionization mass spectrometry technique. Species profiles of key intermediates, products were measured. He et al. [12] investigated the promoting effect using different promoters, including hydrogen, methane, ozone and methanol in a jet-stirred reactor. The results indicated that methanol is the most effective promoter which proves the feasibility of methanol additive. However, there is little intermediate species data available under high pressure, for model validation and kinetic analysis. The trend of NO and N_2O formation under low-intermediate temperatures and high-pressure conditions are still not clear.

In addition, kinetic model for ammonia/methanol consists of pure ammonia, methanol, and interaction mechanism between C and N containing species. Methanol model is well developed and reliable. Burke et al. [17] established a comprehensive model, summarized the previous work and proposed a new one, Mech15.34, based on AramcoMech 1.3 and updated the rate coefficients of methanol unimolecular decomposition reactions and key reactions related to methoxy and hydroxyl-methyl radicals. It can accurately predict measured LBV, IDT and SF data. Our previous work [18] showed that this model can capture the species profile measured under high pressure with great accuracy. Compared with methanol model, there is a relatively large estimation error for ammonia model. We [19] compared the representative ammonia kinetic models published recently using error function value method proposed by Olm et al. [20]. They are not accurate enough to predict the mole fraction of N_2 , NO and N_2O even at atmospheric pressure. Glarborg et al. [21] made a systematic analysis of C-N cross reactions and widely adopted in ammonia model development. But they did not discuss the interactions between ammonia and methanol. Recently, Li et al. [14] conducted theoretical calculations for reactions between amines and species in the methanol subset, supporting the further improvement on ammonia-methanol model. Zhang et al. [22] proposed a new model covering NH_3 /methanol mixture, and the kinetic analysis indicated C-N interactions is less competitive under flame propagation conditions but more competitive under low and

Table 1

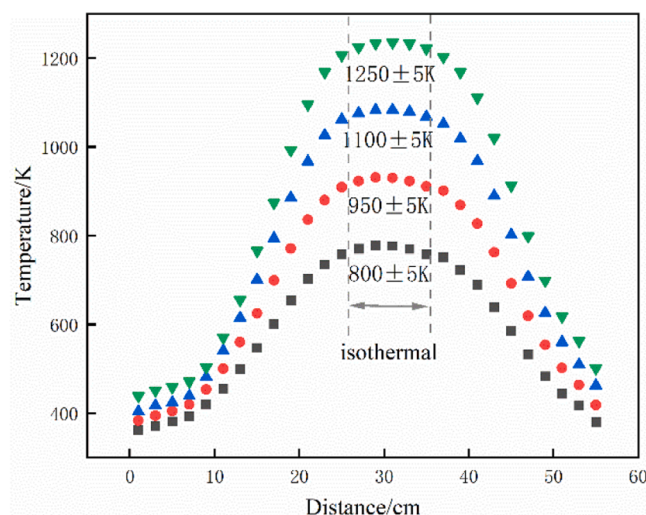
Experimental conditions.

Conditions	Range
Temperature/K	600–1250
Pressure/MPa	5.0
Flow rate/ SCCM	2000
Equivalence ratio	$\phi = 0.5$
Content	NH_3 0.5 %, CH_3OH 0 %, O_2 1.25 %, Ar 98.25 % NH_3 0.495 %, CH_3OH 0.005 %, O_2 1.525 %, Ar 98.2475 % NH_3 0.45 %, CH_3OH 0.05 %, O_2 1.275 %, Ar 98.225 % NH_3 0.375 %, CH_3OH 0.125 %, O_2 1.3125 %, Ar 98.1875 % NH_3 0.25 %, CH_3OH 0.25 %, O_2 1.375 %, Ar 98.125 %

**Fig. 1.** The schematic diagram of the FR used in this study.

intermediate temperature. Due to the lack of species data, the effect of C-N cross reaction on ammonia and methanol consumption and NO_x emission under high pressure and low-to-intermediate-temperature conditions is not yet known.

Therefore, this work addresses the insufficient study encountered in ammonia and methanol oxidation under low-to-intermediate-temperature and high-pressure conditions, such as the shortage of species mole fraction data, C-N interaction mechanism validation and kinetic analysis of fuel consumption and NO_x formation. We measured the key intermediates and products in $\text{NH}_3/\text{CH}_3\text{OH}$ in a flow reactor, at 5.0 MPa, 650–1250 K. A comprehensive kinetic model is proposed based on our previous model work for pure ammonia and methanol oxidation [18,23]. Detailed kinetic analysis was conducted to investigate the dominant reaction pathways involved in ammonia, methanol oxidation and NO_x production, as well as the role of C-N interactions under high pressure and low-to-intermediate-temperature conditions.

**Fig. 2.** Temperature profile of the FR.

2. Experimental methods

Our experiment was carried out in a flow reactor at $\phi = 0.5$, 5.0 MPa and the temperature range from 650 to 1250 K. The detail conditions are listed in Table 1. The fuel mixture consists of ammonia and methanol with the substitution ratio (SR) of methanol varying from 0 to 0.5, where $SR = \frac{V(\text{CH}_3\text{OH})}{V(\text{CH}_3\text{OH}) + V(\text{NH}_3)}$ and $V(\text{CH}_3\text{OH})$ and $V(\text{NH}_3)$ refer to the volume of CH_3OH and ammonia, respectively. Oxygen content is defined based on equivalence ratio with the global reaction definition of $\text{NH}_3 + 1.25\text{O}_2 = \text{NO} + 1.5\text{H}_2\text{O}$.

As depicted in Fig. 1, the flow rates of fuels, oxygen and argon were controlled by mass flow controller. A back-pressure valve was adopted to realize the high pressure. The reactor was designed as a two-layer structure to achieve pressure balance, where exterior is of stainless steel and inner layer is of recrystallized alumina to avoid catalytic reactions. Distribution curve of temperature along the axis in the heating electric stove was measured using a Type K thermocouple under 5 MPa. Fig. 2 shows that the length of constant temperature area was 11–13 cm which can be used to calculate the residence time. The residence time in the FR is calculated according to the formula:

$$\tau = \frac{V}{F} = \pi r^2 L \left(F_N \frac{T}{273.15} \frac{0.1}{P} \right)^{-1}$$

Where τ is residence time/min; V is reaction volume in the constant temperature zone; F is actual flow rate/ $\text{ml}\cdot\text{min}^{-1}$; r refers to inner diameter/cm; L is length of constant temperature zone/cm; F_N stands for standard flow rate/SCCM; T means temperature in the constant temperature zone/K and P is the pressure/MPa.

Species detection can be classified into three categories: 1) The hydrocarbon species concentration was determined by GC/GC-MS through a thermal conductivity detector (TCD) for O_2 , H_2 , CO , CO_2 , N_2 and a flame ionization detector (FID) for methanol and formaldehyde. 2) A chemiluminescence NO_x analyzer was adopted to measure the NO profile. 3) A Fourier transform infrared spectrum (FTIR) is used to quantitatively detected ammonia and N_2O concentrations. The detection limits for NO_x analyzer, TCD, FTIR were 1 ppm, 10 ppm and 10 ppm, respectively, causing 1 % uncertainty for major species like NH_3 , O_2 , and N_2 , and 5 % uncertainty for intermediates like NO and N_2O . Additionally, the random error of the measurement is less than 5 % on the basis of test repeatability. In general, these factors result in an uncertainty of about 10 % for the species. All the experimental data were provided in Supplemental data.

3. Chemical kinetic model

The based ammonia model is from our previous work [23] and methanol is from Burke et al. [17] whose accuracies have been widely validated and can capture the specie profiles measured under high pressure with great accuracy. Additionally, it can accurately predict the measured laminar flame speeds and ignition delay times. Present work pays more attention to the modification on interactions between C- and N-containing species. Key intermediates produced in ammonia oxidation process such as NH_i , NO_x can react with C-containing species. NH_i reacting with alkyl and alkenyl radicals plays an important role in fuel consumption under low-to-intermediate-temperature range [19]. NO is easily oxidized to NO_2 and triggers the formation of OH radical, greatly facilitating the reactivity. Reactions related to NO_2 are favored in fuel oxidation due to its strong oxidizing power. Moreover, C-containing species participate in “ $\text{NO}-\text{NO}_2$ ” looping mechanism and its importance has been emphasized by Shrestha et al. [24] Thus, in this work, four reaction categories are reconsidered including 1) N-containing species (NH_i and NO_x) and CO/H_2 subset, 2) N-containing species and CH_2O subset, 3) N-containing species and CH_2OH subset, 4) N-containing species and CH_3OH .

N-containing species with CO/H_2 subset: Although CO is a stable

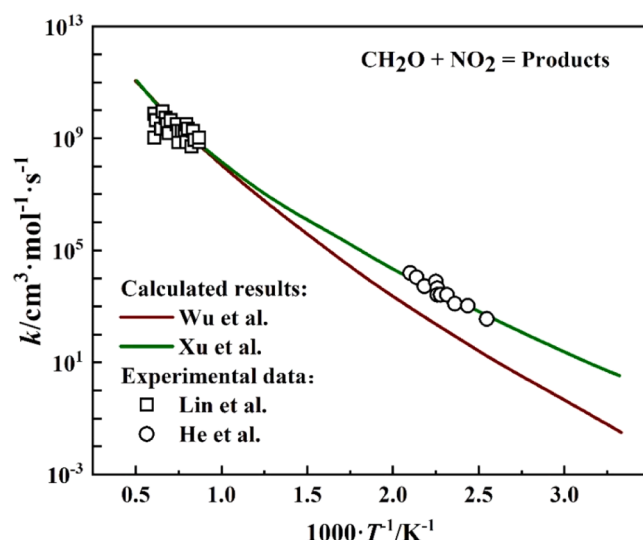


Fig. 3. Comparison between the rate coefficients from Wu et al. [37] and Xu et al. [36] with experimental data.

molecular, NO_2 is very oxidative and can transform CO to CO_2 through R1. This reaction plays an important role in CO_2 formation in ammonia and C-containing species co-firing. Kroupnov and Pogosebikian [25] calculated the potential energy surface of NO_2+CO system at B3LYP/cc-pVTZ level, and estimated the rate coefficients using transition state theory at 300–2500 K, which was adopted in Zhang model. [26] However, at lower temperature, the above results are 30 times higher than the measured data proposed by Johnston et al. [27] and Brown et al. [28] Later, Sun et al. [29] recalculated the rate coefficients in NO_2+CO system at CCSD(T)/CBS level. The results agree well with the measured data under the overall temperature range. Consequently, results from Sun et al. [29] were adopted in our present work.

Additionally, N_2O is one of the major emissions under intermediate temperature and high-pressure conditions. Korasev et al. [30] found that N_2O consumption is sensitive to reaction $\text{N}_2\text{O}+\text{H}_2 = \text{N}_2+\text{H}_2\text{O}$ (R2) and estimated the rate coefficient by fitting experimental data. Recently, according to the modeling work on $\text{NO}_x/\text{H}_2/\text{CO}$ from Kovács et al. [31], there is a relatively large error for data from Korasev et al. [30] Repeated experiment and rate coefficient estimation were conducted by Mulvihill et al. [32] and Sun et al. [29], respectively. The results from Sun et al. [29] showed that through reducing the rate coefficient by a factor of 5, their computed results agree well with the measured data in Mulvihill et al. [32] Therefore, it was incorporated in present model.



HCO is a dominant intermediate radical in CH_3OH oxidation. High level quantum calculation results for reactions between HCO and HNO or HONO from Xu [33] were added in this work. The interaction mechanism between HCO and NO_x is the key to DeNO $_x$ channels. Xu et al. [34] investigated the H atom abstraction reaction of HCO by NO using ab initio methods and they found R5 is the most favorable channel, accounting for more than 90 % branching ratio. Their result agrees well with the measured rate coefficients by Dammeier et al. [35] over 769–1307 K and was adopted in current work. Dammeier et al. [35] measured the rate coefficients of HCO with NO_2 system in a flow reactor and a shock tube at the intermediate temperature range from 298 K to 1050 K. They estimated the branching ratio of three products channels (R6–R8) and derived the rate coefficients.





N-containing species with CH₂O subset: CH₂O is one of the dominant species and emissions generated in methanol oxidation. The mechanism between N-containing species with CH₂O was investigated. Xu and Lin [36] calculated the potential energy surface and rate coefficients of CH₂O+NO₂ at MPW1PW91/6-311+G(3df, 2p)/B3LYP/6-311+G(d,p) level. The results showed that HONO+HCO and HNO₂+HCO are the dominant bimolecular product channels for this system. In addition, high level quantum calculation was also obtained by Wu et al. [37] Thus, we compared their calculated rate coefficient of R9 with experimental data as presented in Fig. 3. Both results agree well with data at high temperature while only rate coefficient calculated by Xu et al. [36] matches well with low temperature data. Therefore, we adopted the theoretical calculations in Xu et al. [36] The work on H abstraction reaction from CH₂O by NH₂ (R11) is still scarce, only Li et al. [38] theoretically calculated this reaction and was added in present work.



N-containing species with CH₂OH/CH₃O subset: CH₂OH and CH₃O are largely produced through H elimination reactions of methanol, thus the rate coefficients of these two intermediates with NO_x or NH₂ (R12-R20) are revised. For CH₂OH+NO, HNCO + H₂O is the dominant product which is proved in Feng et al. [39] by infrared diode laser spectroscopy. Feng et al. [39] also proposed the branching ratio of this bimolecular product channel. Therefore, in present work, the global rate coefficient of R12 in Nesbitt et al. [40] together with the branching fraction from Feng et al. [39] was adopted in the present model. The reactions between CH₂OH and NO₂ were explored by Zhang et al. [41] theoretically using electronic structure methods and transition state theory. They found that CH₂O+HONO is the most dominant bimolecular product. The next most competitive bimolecular product is CH₂O + HNO₂. The global rate coefficient from Nesbitt et al. [40] together with the branching fractions from Zhang et al. [41] were adopted in present work. Rate coefficients of CH₂OH or CH₃O with NH₂, i.e. R15-R20, were from the auto-estimated results in Dean et al. [42] by RMG.



N-containing species with CH₃OH subset: The interaction mechanism between CH₃OH and NO_x mainly includes R21-R24. The potential energy diagram was obtained at CCSD(T)/aug-ccpVQZ//M06-2X-D3/6-311++G(d,p) in Wu et al. [37] and their computed global rate constant matches well with the previous measurements. Therefore, we adopted

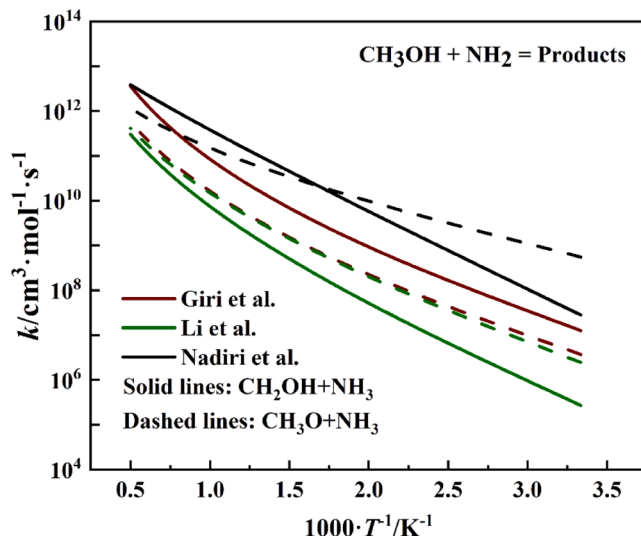


Fig. 4. Comparison between the rate coefficients from Giri et al. [43], Li et al. [14] and Nadiri et al. [15].

Table 2

The rate coefficients of the modified reactions in this work in Arrhenius expression $k = AT^n \exp(-E_a/(RT))$. Units are mol, cm, s, cal.

	Reactions	A	n	E _a	Source
R1	CO+NO ₂ = CO ₂ +NO	3.9E01	3.310	26221	[29]
R2	N ₂ O+H ₂ = N ₂ +H ₂ O	3.5E13	0.000	32504	[29]
R3	HCO+HNO = CH ₂ O+NO	5.8E-1	3.840	115	[33]
R4	HCO+HNO = H ₂ NO+CO	4.9E01	3.270	1754	[33]
R5	HCO+NO = HNO+CO	1.03E10	1.470	-1764	[34]
R6	HCO+NO ₂ = NO+CO ₂ +H	2.31E10	0.000	0	[35]
R7	HCO+NO ₂ = HONO+CO	4.95E12	0.000	0	[35]
R8	HCO+NO ₂ = OH+NO+CO	4.95E12	0.000	0	[35]
R9	CH ₂ O+NO ₂ = HONO+HCO	1.4E-07	5.640	9220	[36]
R10	CH ₂ O+NO ₂ = HNO ₂ +HCO	1.07E-1	4.220	19850	[36]
R11	CH ₂ O+NH ₂ = HCO+NH ₃	6.3E04	3.000	3770	[38]
R12	CH ₂ OH+NO = HNCO+H ₂ O	1.0E11	0.000	0	[40]
R13	CH ₂ OH+NO ₂ = CH ₂ O+HONO	5.0E12	0.000	0	[40]
R14	CH ₂ OH+NO ₂ = CH ₂ O+HNO ₂	-	-	-	[21]
R15	CH ₂ OH+NH ₂ = CH ₂ O+NH ₃	1.3E06	1.940	1150	[44]
R16	CH ₂ OH+NH ₂ = NH+CH ₃ OH	5.6E06	1.940	14573	[44]
R17	CH ₂ OH+NH ₂ = HCOH+NH ₃	3.2E06	1.870	7570	[44]
R18	CH ₂ OH+NH ₂ = CH ₂ NH ₂ +OH	1.0E12	0.390	3497	[42]
R19	CH ₃ O+NH ₂ = CH ₂ O+NH ₃	3.3E06	1.940	-1150	[44]
R20	CH ₃ O+NH ₂ = H ₂ NO+CH ₃	6.8E14	0.110	2332	[42]
R21	CH ₃ OH+NO ₂ = CH ₂ OH+HONO	2.4E06	2.050	25160	[26,37]
	DUP	3.9E05	1.490	18560	
	DUP	5.6E-2	4.180	26320	
	DUP	1.6E-4	4.830	83900	
R22	CH ₃ OH+NO ₂ = CH ₂ OH+HNO ₂	1.8E01	3.500	23240	[26,37]
	DUP	2.9E-1	0.560	16480	
R23	CH ₃ OH+NO ₂ = CH ₃ O+HONO	2.1E-3	0.210	25030	[26,37]
	DUP	1.2E01	3.510	25810	
	DUP	1.1E-1	0.340	33770	
	DUP	9.6E-4	4.600	28000	
R24	CH ₃ OH+NO ₂ = CH ₃ O+HNO ₂	1.1E02	3.270	30000	[26,37]
	DUP	1.7E-1	1.700	39240	
R25	CH ₃ OH+NH ₂ = CH ₂ OH+NH ₃	4.97E-2	4.300	3086	[43]
R26	CH ₃ OH+NH ₂ = CH ₃ O+NH ₃	5.20E-2	4.050	2992	[43]

this high-level quantum calculation results in present model.



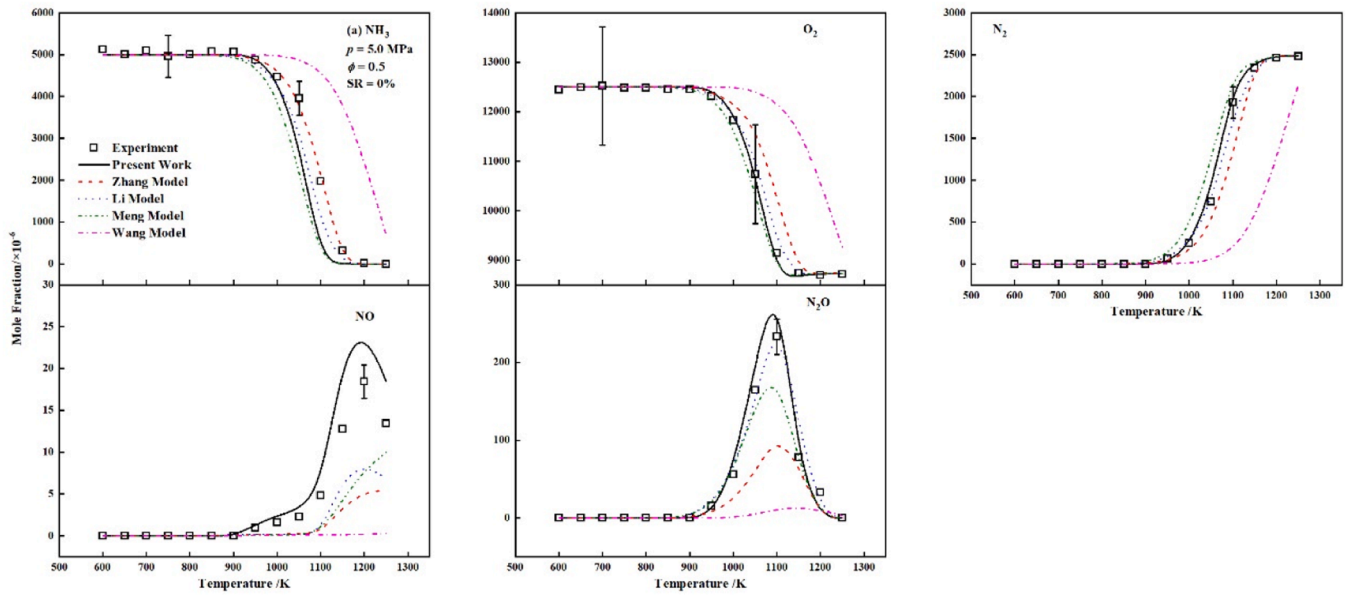
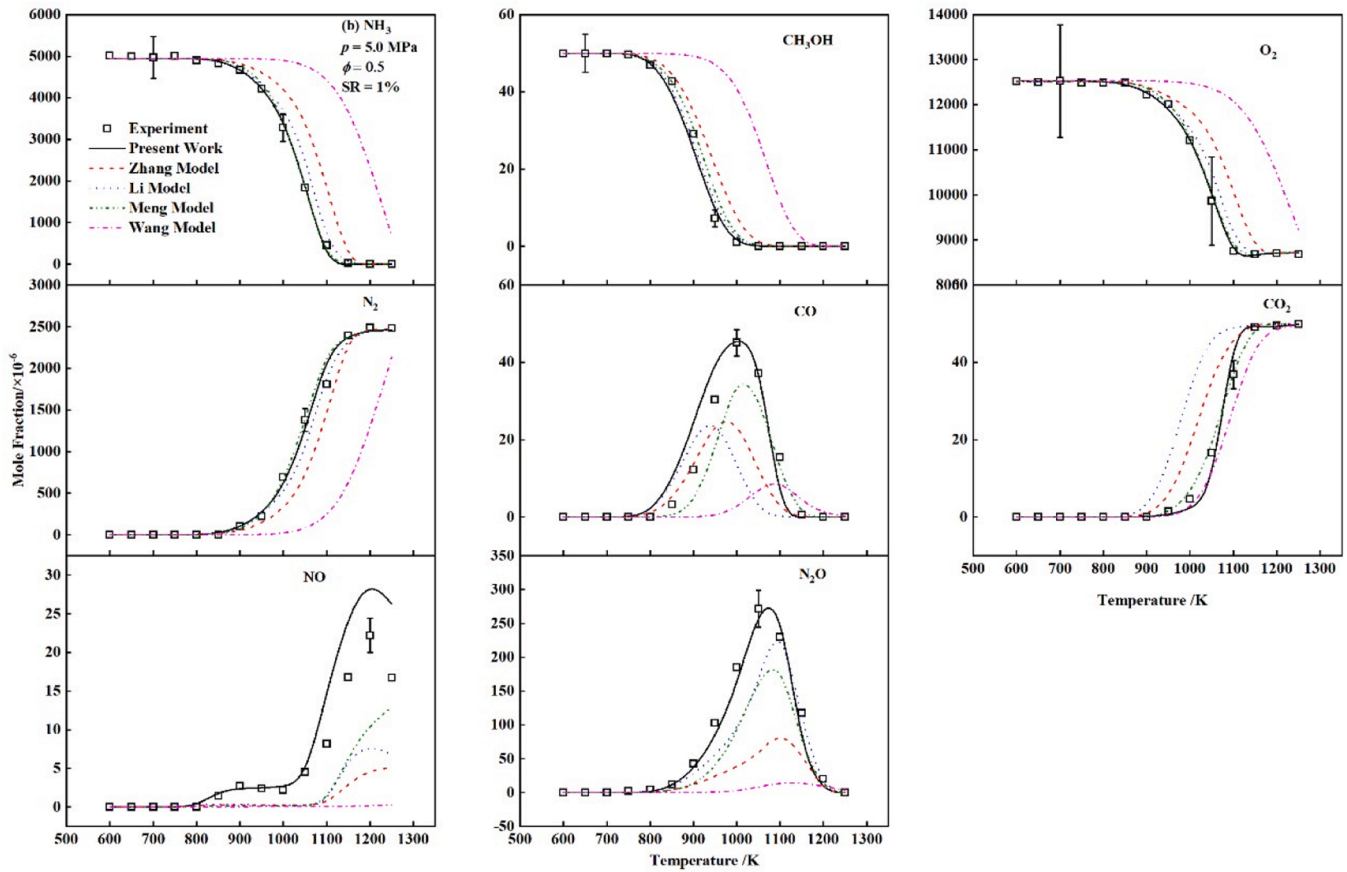
(a) $SR = 0\%$ (b) $SR = 1\%$

Fig. 5. Measured (symbols) and simulated (lines) mole fraction profiles using different models of ammonia and methanol oxidation in FR at 5.0 MPa, $\phi = 0.5$, under different methanol blending ratio.

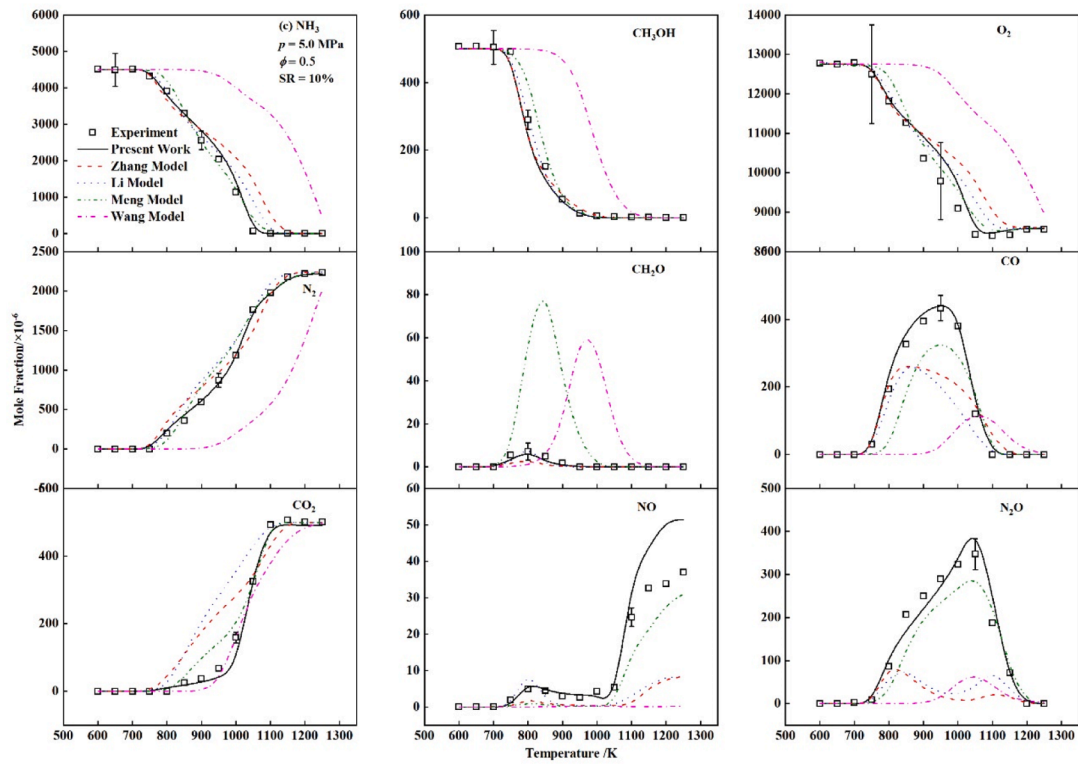
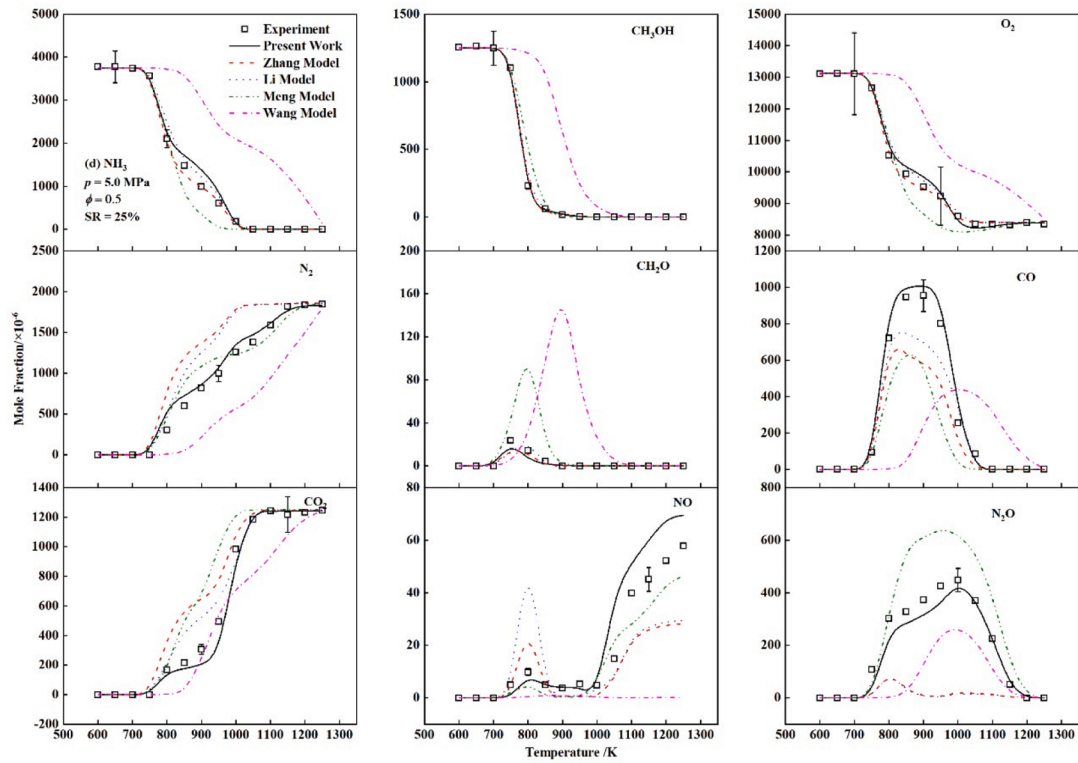
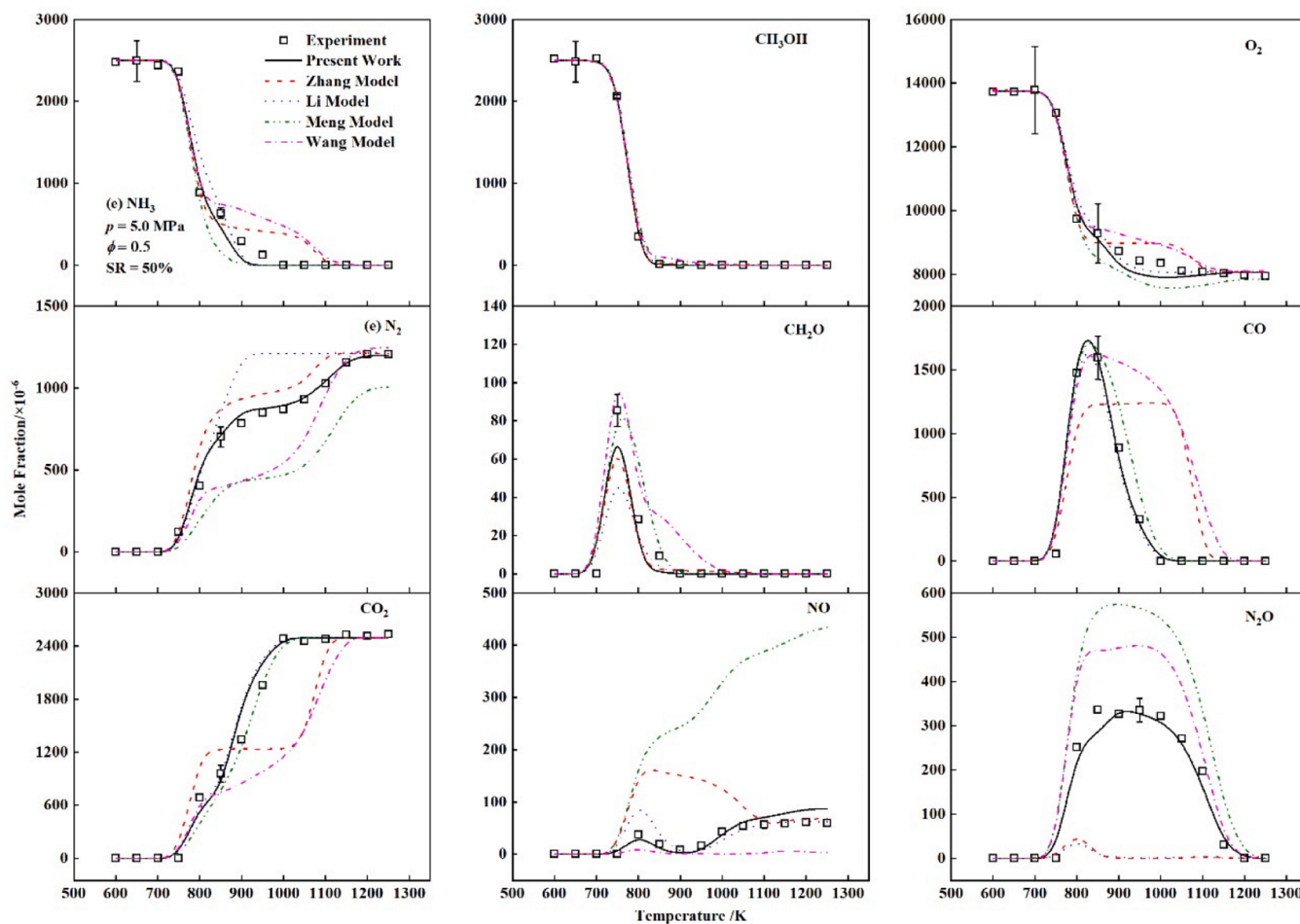
(c) $\text{SR} = 10\%$ (d) $\text{SR} = 25\%$

Fig. 5. (continued).



(e) SR = 50%

Fig. 5. (continued).

Reactions between CH_3OH and NH_2 take the dominant place in IDT and LBV of ammonia-methanol mixtures [14]. NH_2 can abstract H atom from methyl or hydroxyl in methanol as listed in R25-R26. Giri et al. [43] theoretically investigated the kinetics of $\text{CH}_3\text{OH} + \text{NH}_2$ at CCSD (T)/CBS//M06-2X/aug-cc-pVTZ, and found R25 is more favorable with lower barrier height accounting for 83.7 % of the total flux. Li et al. [14] also calculated the rate coefficients of the same reaction class. However, an opposite trend was observed that H atom abstraction from hydroxyl (R26) is more dominant as presented in Fig. 4. He et al. [12] compared these two results and found calculated rate coefficients from Giri et al. [43] can provide more accurate predictions on IDT data, as the results from Giri et al. [43] appear to be more consistent with the general rules of H elimination in CH_3OH . Therefore, this result was adopted in present model. The rate coefficients revised in present work were listed in Table 2. The present model is provided as Supplemental data.



4. Results and discussion

4.1. Measured species profiles and model validation

Fig. 5 depicts the measured mole profiles during ammonia-methanol oxidation process and compared with the simulated results using different models. The initial reaction temperature of ammonia decreases with methanol ratio. H_2 and N_2 are key products with mole fractions higher than 1000 ppm. Similarly, an enormous number of CO and CO_2 are produced in methanol oxidation. As to emission, NO and N_2O are generated and change non-monotonically with temperature, especially under higher methanol ratio conditions. NO first increases then decreases, and finally increases with temperature. With the increase of temperature, N_2O first rises and then falls down. CH_2O is only produced at 700–900 K temperature range and can be entirely consumed at the temperature higher than 900 K.

The elemental balance at SR=25 % experimental condition of C and N as shown in Fig. 6. As can be seen from the figure, the C equilibrium error reaches 10 % at $T = 950$ K due to the vigorous reaction, and is only at 1 % at 650 K at the beginning of the reaction and at 1200 K when CH_3OH is completely consumed.

Regarding model validation, present model is compared with the recently proposed ammonia/methanol model, including Zhang et al. [22], Li et al. [14], Meng et al. [45] and Wang et al. [13], as these mechanisms can provide good predictions for the laminar flame speeds

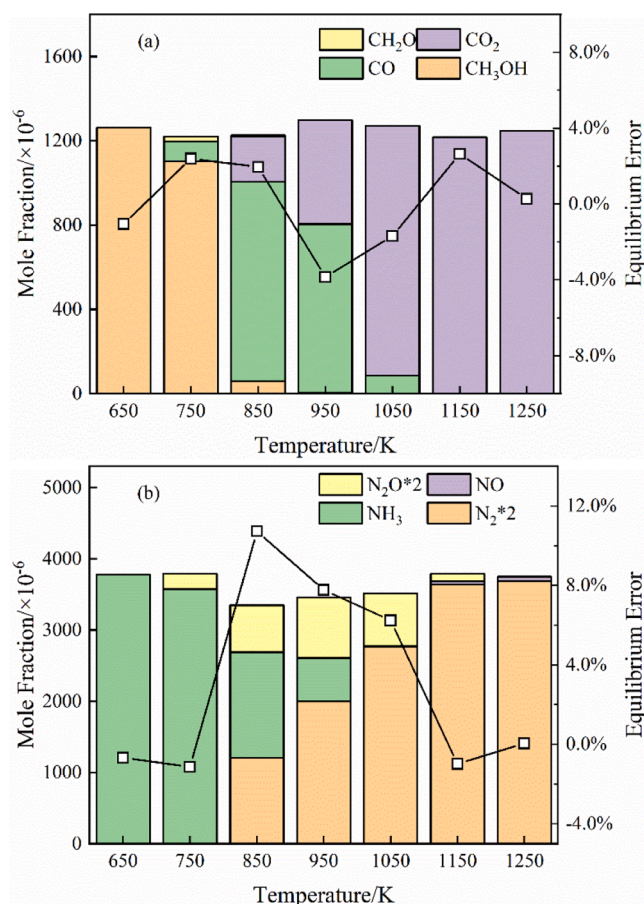


Fig. 6. Elemental balance at SR=25 % experimental condition of (a) C and (b) N.

or ignition delay times for ammonia and methanol mixture. Fig. 5 shows the comparison between the measured data and simulated results during ammonia/methanol oxidation process. Among all the model, Wang model [13] underpredicts the overall reactivity under wide conditions. Ammonia and methanol are rapidly consumed at around 800 K, at SR = 0.25. However, the simulated reaction temperature using Wang model [13] is 900 K, much higher than the actual temperature. Oppositely, Meng model [45] exhibits a higher oxidation rate of fuel mixture at the methanol blending ratio bigger than 25 %. Other models, including Zhang model [22], Li model [14] and our present model can accurately predict the concentration profiles of NH_3 , O_2 , CH_3OH and CO_2 . But only present model can accurately capture the non-monotonical trend of NO and N_2O under current conditions. In general, the present model can adequately predict the species concentration during ammonia/methanol oxidation. Compared with other mechanisms, the present mechanism has improved the prediction accuracy of the NOx concentrations under wide conditions. All the measured data are listed in Supplemental data.

Model validation against the speciation profiles of fuels, intermediates and products during ammonia-methanol mixtures oxidation from He et al. [12] was conducted. Fig. 7 presents the measured mole fraction versus temperature of species including NH_3 , CH_3OH , NO, N_2 , CO and CO_2 compared with Present model, Zhang model [22], Li model [14], Meng model [45] and Wang model [13]. Wang model [13] apparently underpredicted the overall reactivity under current condition and cannot capture the trend of all species. Other models can capture the trends of CO and CO_2 . Zhang model [22] and Present model can accurately predict the measure mole fractions of NH_3 , CH_3OH , and N_2 , while Present model has better predictions on measured NO mole fractions.

Ignition delay time helps understand the oxidation mechanism at intermediate and high temperature range. Model validation against the measured ignition delay times is presented in Fig. 8. Experimental data was obtained in a shock tube with varying the methanol content at different equivalence ratios and pressures. Similarly, Wang model [13] apparently underpredicted the overall reactivity under current condition and provides much higher ignition delay times than others. Other models, especially the Present model and Li model [14] can provide accurate predication on ammonia and methanol mixture ignition under the majority conditions. But all the model overpredicts the ignition delay times, at low temperature, 30 % methanol blending. The deficiency of the models in predicting the ignition delay times is mainly due to the underestimating the reactivity of methanol.

Laminar flame speed data were also selected to test our model. Fig. 9 presents the comparison between the measured data and simulated results from the Present model, Wang model [13] and Konnov model [46], while Wang [13] and Konnov model [46] were validated against the measured laminar flame speeds of ammonia mixed with methanol in Wang et al [13]. The present model can reasonably predict the experimental data under different methanol blending ratios.

4.2. $\text{NH}_3/\text{CH}_3\text{OH}$ oxidation and Effects of methanol ratio on NH_3 conversion

Fig. 10 depicts the measured mole profiles of ammonia and methanol at different methanol blending ratio compared with the simulated results using the present model. The mole fractions of methanol were normalized and presented in the same figure. With more methanol addition, the ammonia's onset oxidation temperature decreases from 1000 K to around 750 K. There is a similar trend of methanol concentration variation. Kinetic analysis was conducted which focused on the key reaction pathways of ammonia oxidation varying with methanol blending ratio at high pressure. More attention was paid for the interaction mechanism of C- and N-containing species, and their effect on ammonia oxidation.

The bond dissociation energies of C-H bond in methanol and N-H bond in ammonia are 96.08 kcal/mol and 107.57 kcal/mol, respectively. Thus, the chain initial reaction of methanol can be triggered at lower temperature. During the ammonia-methanol oxidation process, H atom is firstly abstracted from methanol by oxygen, producing HO_2 and further yielding H, OH and O radicals, through chain branching or propagation reaction. These reactions also trigger the oxidation of ammonia. Based on the modified model, sensitivity analysis was conducted at the ammonia conversion rate of 10 % ($\omega(\text{NH}_3) = 10\%$) as shown in Fig. 11. The positive sensitivity coefficient denotes repression on ammonia oxidation.

Under pure ammonia oxidation condition, $\text{H} + \text{O}_2(+\text{M}) = \text{HO}_2$ exhibits a promoting role in ammonia oxidation especially under high pressure as its rate coefficient is strengthened. HO_2 can react with NH_3 producing H_2O_2 with H_2O_2 quickly decomposing to OH radical and facilitating the ammonia oxidation. The chain propagation reaction $\text{NH}_2 + \text{NO} = \text{NNH} + \text{OH}$, shows a high negative sensitivity coefficient, as the OH production can promote the ammonia consumption. $\text{NH}_2 + \text{NO} = \text{N}_2 + \text{H}_2\text{O}$ exhibits positive sensitivity coefficient with stable N_2 and H_2O produced and it also competes with OH producing channel. Similarly, $\text{NH}_2 + \text{NO}_2 = \text{H}_2\text{NO} + \text{NO}$ and $\text{NH}_2 + \text{NO}_2 = \text{N}_2\text{O} + \text{H}_2\text{O}$ are two competitive reactions of a high negative sensitivity and a positive sensitivity coefficient, respectively. It can be found that pure ammonia oxidation is closely correlated with NH_3 or NH_2 reactions. With the addition of methanol, a noticeable reduction in the sensitivity coefficients of ammonia-related reactions is observed as the decreasing importance of ammonia in the oxidation process, whereas promotion reactions related to methanol are enhanced significantly. The OH production is independent of reaction $\text{NH}_2 + \text{NO} = \text{NNH} + \text{OH}$, but related to the new elementary reaction $\text{CH}_3\text{OH} + \text{HO}_2 = \text{H}_2\text{O}_2 + \text{CH}_2\text{OH}$. Thereby, the latter becomes the most dominant reaction in promoting the ammonia

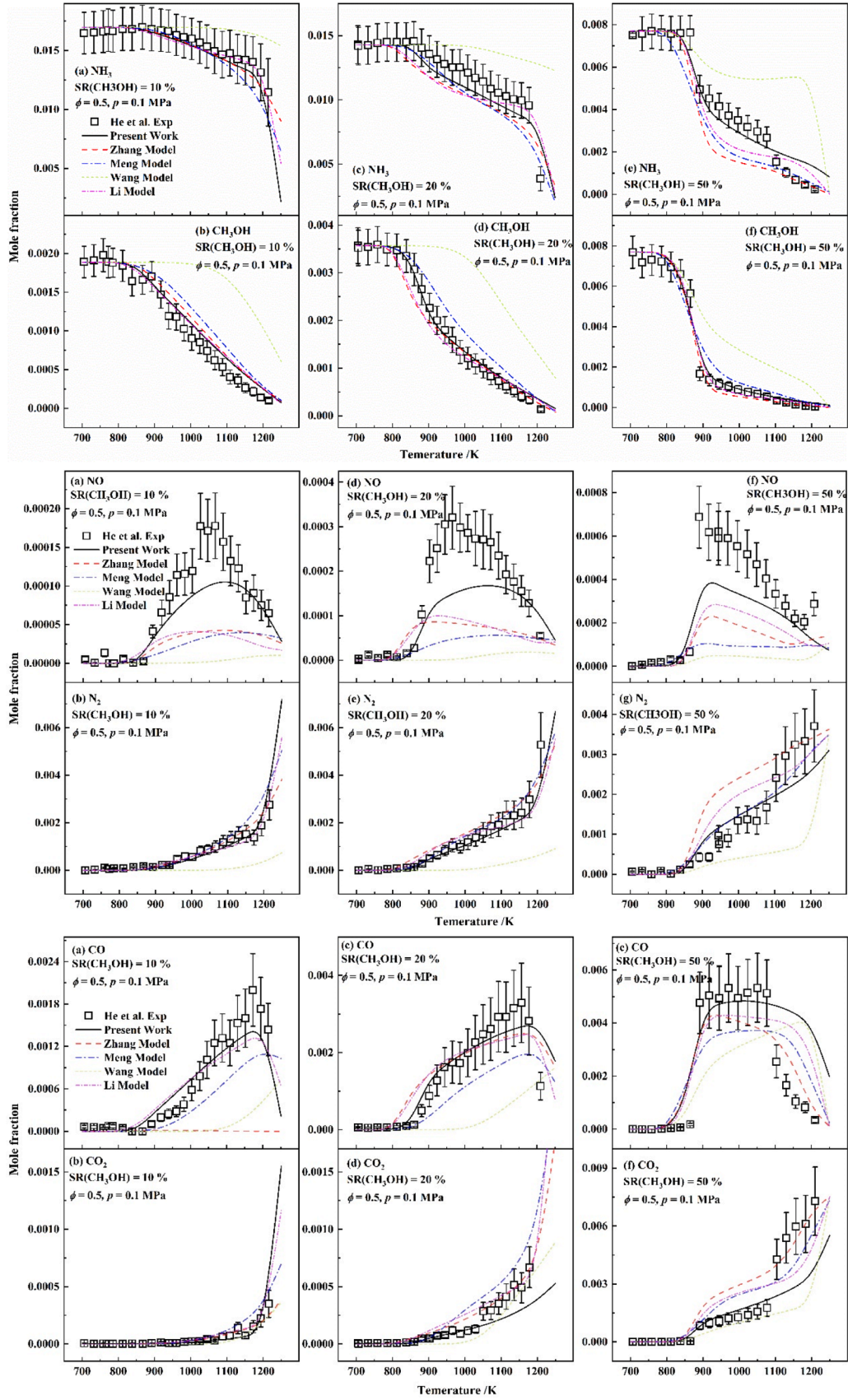


Fig. 7. Measured (symbols) and simulated (lines) mole fraction profiles using different models of ammonia and methanol oxidation in a jet stirred reactor at 0.1 MPa, $\phi = 0.5$, under different methanol blending ratio.

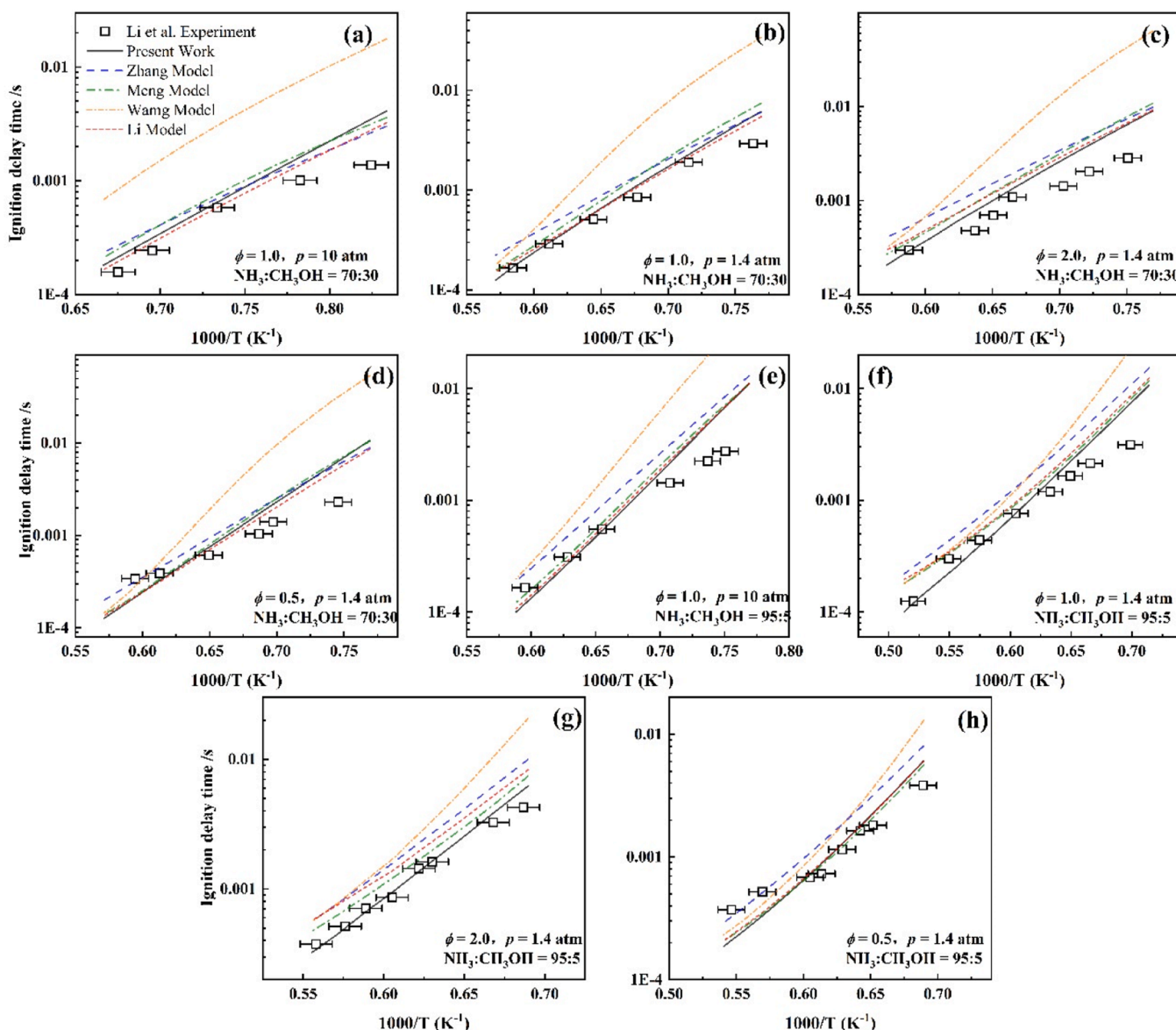


Fig. 8. Measured (symbols) and simulated (lines) ignition delay times using different models of ammonia and methanol oxidation in a shock tube.

oxidation. A shift in the sequence of important reactions can be observed. Most notably, the cross reactions between C- and N-containing species, $\text{CH}_2\text{O} + \text{NH}_2 = \text{HCO} + \text{NH}_3$ and $\text{CH}_3\text{OH} + \text{NH}_2 = \text{CH}_2\text{OH} + \text{NH}_3$ play a promoting role in reactivity at 1 % methanol addition, but an inhibit role at 10 % methanol blending. Radical pool and pathway analysis will help us further explore the deep insight.

As discussed above, HO_2 and OH can be produced from methanol related reactions and thus changes the sequence of important reactions. To be more clearly, Fig. 12 shows the peak mole fractions of OH , HO_2 , NH_2 , HNO and H_2NO . HO_2 is majorly produced at low temperature and high pressure and substantial increases with increasing methanol addition. Similar trend can be observed for OH , HNO and H_2NO radicals. Although methanol addition promotes NH_3 oxidation but C-N interaction would transform NH_2 to NH_3 at the same time. In addition, the dilution effect reduces the NH_2 production. Thus, NH_2 concentration increases with methanol blending slightly.

Fig. 13 shows the pathway diagram of ammonia oxidation at 1000 K, 5.0 MPa, $\phi = 0.5$ and 10 % methanol blending. Ammonia-methanol mixture oxidation can be divided into ammonia oxidation routes (black arrows), methanol oxidation pathway (green arrows), HO_2 and OH formation ways (blue arrows) and C-N interaction mechanism (yellow

arrows). Without methanol addition, ammonia is consumed following the two reaction routes: $\text{NH}_3\text{-NH}_2\text{-NNH-N}_2$ and $\text{NH}_3\text{-NH}_2\text{-H}_2\text{NO-HNO-NO-NO}_2$. Special attention has been paid for the significant effect of methanol on ammonia consumption routes. Firstly, the C-N interaction reactions as the yellow arrows presented, the H-elimination reactions of CH_3OH and CH_2O by NH_2 promotes the consumption of methanol contributing to the formation of CH_2OH and HCO radicals, which further boosts the whole reactivity. However, on the other hand, these two reactions transform NH_2 to NH_3 at the same time which exhibits an inhibition effect. Therefore, $\text{CH}_2\text{O} + \text{NH}_2 = \text{HCO} + \text{NH}_3$ and $\text{CH}_3\text{OH} + \text{NH}_2 = \text{CH}_2\text{OH} + \text{NH}_3$ play a promoting role in reactivity at 1 % methanol addition, but an inhibit role at 10 % methanol blending. Secondly, methanol can produce free radicals, OH and HO_2 during its consumption and facilitating ammonia oxidation. OH radical is produced through $\text{CH}_3\text{OH} + \text{HO}_2 = \text{CH}_2\text{OH} + \text{H}_2\text{O}_2$ and $\text{H}_2\text{O}_2(+\text{M}) = 2\text{OH}(+\text{M})$. Additionally, methanol oxidation accelerates the HO_2 formation through $\text{HCO} + \text{O}_2 = \text{CO} + \text{HO}_2$, and $\text{CH}_2\text{OH} + \text{O}_2 = \text{CH}_2\text{O} + \text{HO}_2$.

4.3. NO_x formation and consumption in $\text{NH}_3/\text{CH}_3\text{OH}$ oxidation process

As shown in Fig. 5, NO and N_2O are generated and change non-

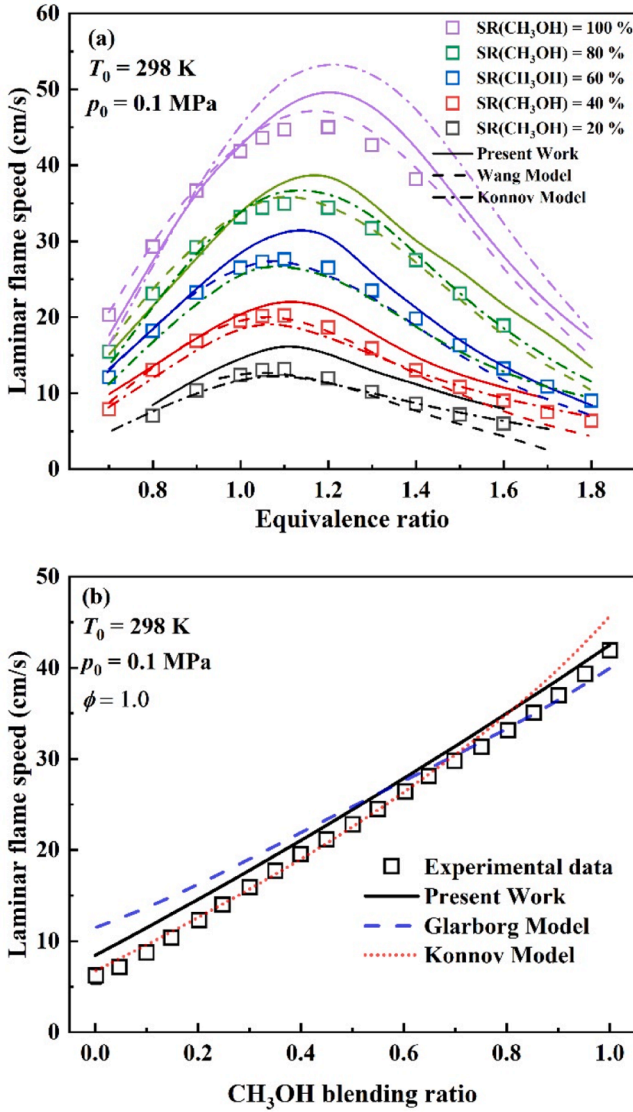


Fig. 9. Measured (symbols) and simulated (lines) laminar flame speeds using Present model, Wang model [13] and Konnov model [46] of ammonia and methanol oxidation in a shock tube.

monotonically with temperature, especially under higher methanol ratio

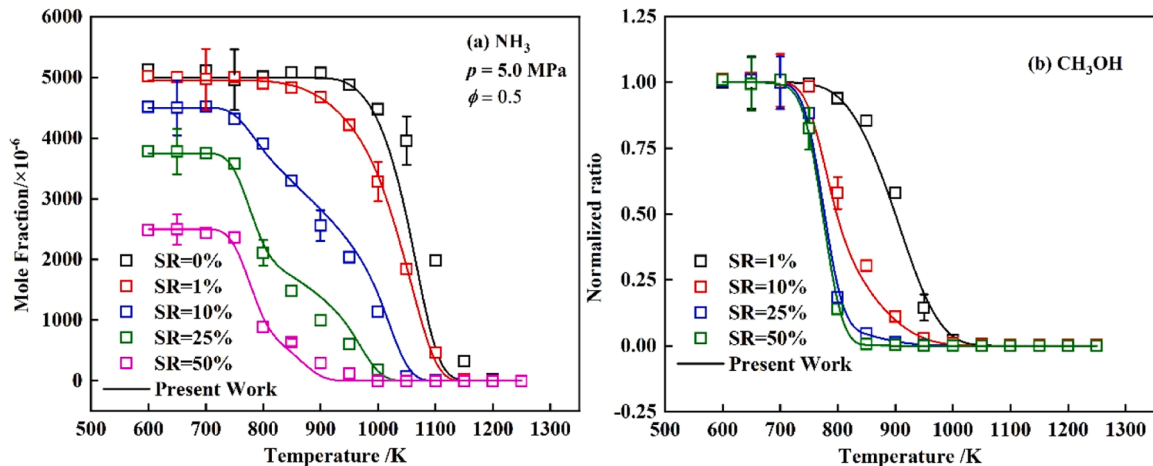


Fig. 10. Measured (symbols) and simulated (lines) mole fraction profiles of ammonia and methanol at 5.0 MPa, $\phi = 0.5$, under different methanol blending ratio.

conditions. NO first increases then decreases, and finally increases with temperature. As to N_2O , it first increases and then decreases with temperature. It is worth noting that there is a trade-off between NO and N_2O at intermediate temperature range. NO increases as N_2O begins to decrease with temperature. To better understand the kinetic insight, pathway analysis was conducted at $T = 750, 800$ and 850 K, referring to increase-decrease trend, as well as 950 K and 1150 K, referring to the further monotonical increase trend.

NO involved pathways can be divided into NO formation and NO reduction reactions. Under current condition, the thermal NOx formation is negligible due to the relatively low temperature. On the other hand, fuel NOx is crucial during ammonia oxidation process. As shown in Fig. 14, the formation of NO is mainly through two routes: (1) $\text{NH}_3 \rightarrow \text{NH}_2 \rightarrow \text{H}_2\text{NO} \rightarrow \text{HNO} \rightarrow \text{NO}$, (2) $\text{NH}_3 \rightarrow \text{NH}_2 \rightarrow \text{NO}$. NH_2 produced from H abstraction from NH_3 , can react with NO_2 forming H_2NO , with H_2NO further reacting with NO_2 , generating HNO or HONO. HNO is transformed directly to NO or HONO. The majority of HONO successively undergoes decomposition to yield NO. H_2NO becomes chain carrier that converts NH_2 to NO. Or, NH_2 combines NO_2 and directly produces NO. The reduction of NOx is mainly through $\text{NH}_2 + \text{NO} = \text{NNH} + \text{OH}$, $\text{NH}_2 + \text{NO} = \text{N}_2 + \text{H}_2\text{O}$ and $\text{NH}_2 + \text{NO}_2 = \text{N}_2\text{O} + \text{H}_2\text{O}$. At 750 K, little

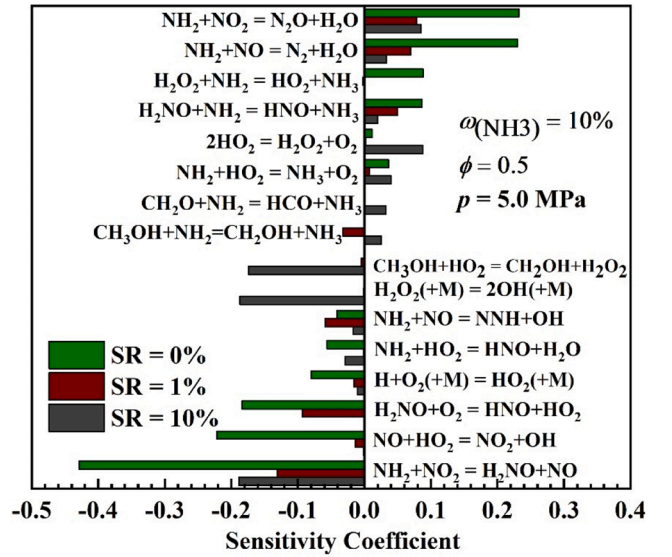


Fig. 11. Sensitivity analysis of ammonia oxidation at 5.0 MPa, $\phi = 0.5$, and at the ammonia conversion rate of 10 % under different methanol blending ratio.

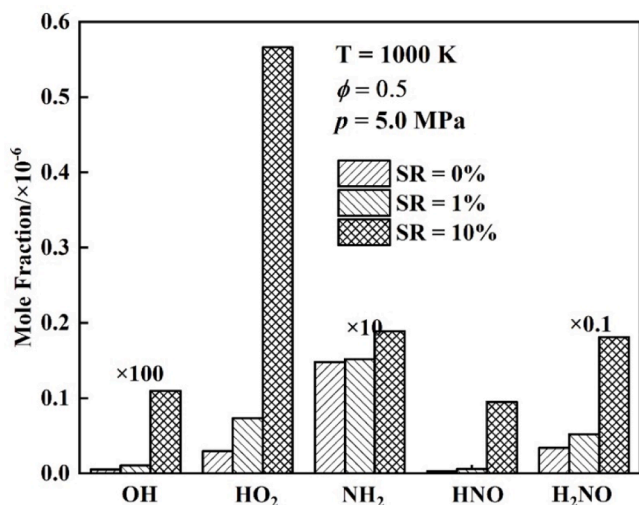


Fig. 12. The peak mole fraction of OH, HO₂, NH₂, HNO and H₂NO at 5.0 MPa, $\phi = 0.5$, and $T = 1000$ K under different methanol blending ratio.

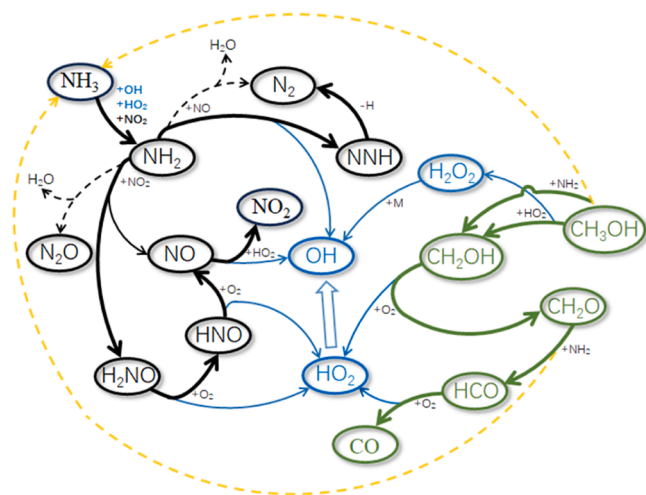


Fig. 13. Pathway diagram of ammonia-methanol oxidation process at 1000 K, 5.0 MPa, $\phi = 0.5$ and 10 % methanol blending. Black arrows referring to ammonia oxidation routes, green arrows referring to methanol oxidation pathway, blue arrows referring to HO₂ and OH formation ways and yellow arrows referring to C-N interaction mechanism.

ammonia is oxidized, accounting for 4.7 % of the total flux and thus NO is generated in small amounts. With temperature increasing to 800 K, NH₃ is rapidly consumed with a flux up to 48.1 % and generates a large amount of NH₂ radical. According to the pathway diagram, NO formation through NH₂ is favorable than NO reduction. Therefore, NO concentration increases quickly as temperature increases. At 850 K, the NO reduction pathway proportion is enhanced from 21.9 % to 35.8 % with the increase of temperature. They compete with NO formation routes, leading to the gradually decrease of the NO mole fraction.

The formation route of N₂O is relatively simple, which mainly goes through NH₂ + NO₂ = N₂O + H₂O. As the temperature is increased from 750 K to 850 K, the cumulative flux of N₂O generation increases from 3.4 % to 9.6 %, leading to monotonical increase of N₂O concentration.

Fig. 15 depicts NO formation related pathways (left part) and NO reduction (right part) at 950 K (black) and 1100 K (red). At current operating temperature range, ammonia has been totally consumed which means that NH₂ concentration is temperature independent. In addition, the branching ratios of NO formation pathways almost keep unchanged (35.7 % and 35.9 % of the total flux) while the flux through

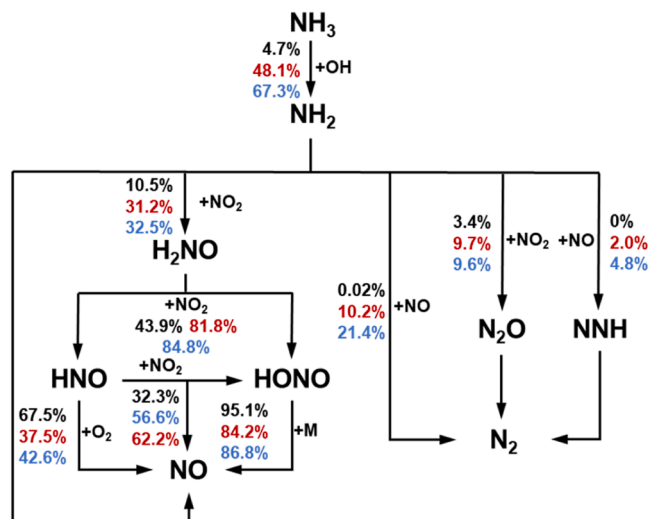


Fig. 14. Pathway analysis of ammonia-methanol oxidation process at $T = 750$ (black), 800 (red) and 850 K (blue), 5.0 MPa, $\phi = 0.5$ and 10 % methanol blending.

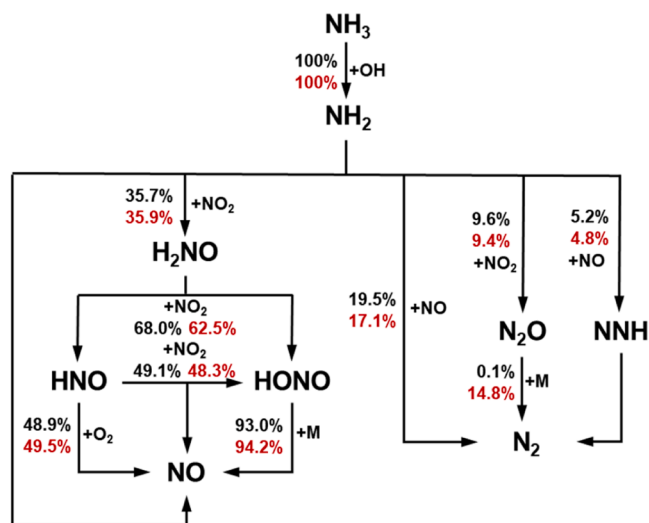


Fig. 15. Pathway analysis of ammonia-methanol oxidation process at $T = 950$ (black), and 1100 (red), 5.0 MPa, $\phi = 0.5$ and 10 % methanol blending.

the DeNOx channel drops from 34.3 % to 31.3 %. Moreover, the increasing temperature boosts the generation of oxygenated radicals like O, OH and HO₂. These radicals make NO formation reactions more favorable than NO reduction reactions, leading to the enhancement of NO formation.

As to N₂O formation, NH₂ radical undergoes NH₂ + NO₂ = N₂O + H₂O with almost the same branching ratios (9.6 % and 9.4 % of the total flux) at different temperature. In addition, the reaction N₂O = N₂ + O becomes dominant. The flux through the decomposition of N₂O increases from 0.1 % to 14.8 %, leading to a decrease tendency.

4.4. Effects of methanol ratio on NOx formation and consumption

To better compared the NOx formation trend, the measured and predicted mole fractions of NO and N₂O versus temperature at different methanol blending ratio are illustrated in Fig. 16. It is found that the non-monotonical trend is enhanced by methanol addition. The peak mole fraction of NO at the temperature range of 700–900 K increases with more methanol blending. In addition, at 1200 K, NO concentration

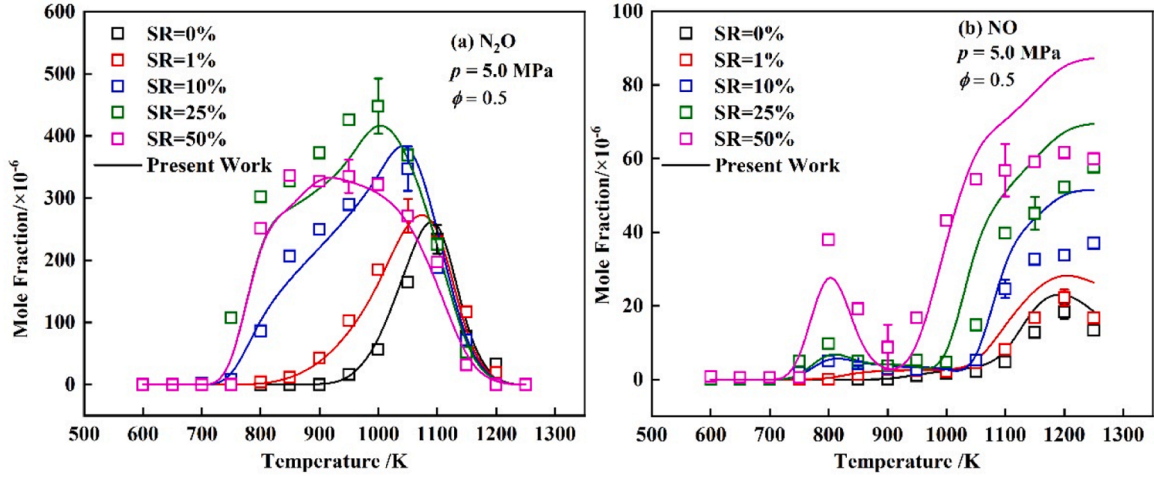


Fig. 16. Measured (symbols) and simulated (lines) mole fraction profiles of NO and N_2O at 5.0 MPa, $\phi = 0.5$, under different methanol blending ratio.

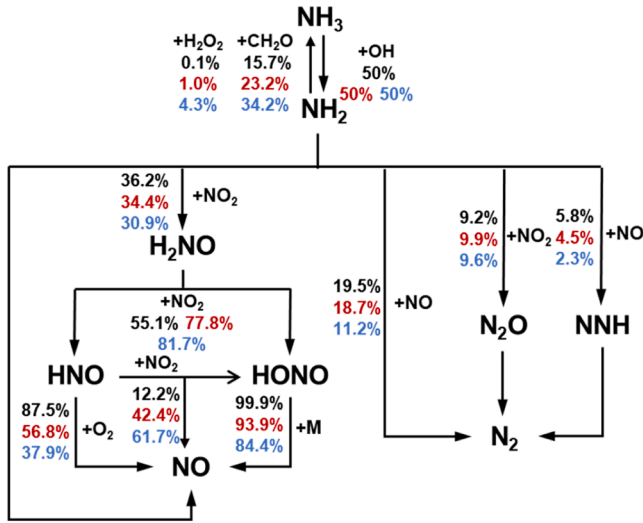


Fig. 17. Pathway analysis of NOx formation and reduction at 10 (black), 25 (red) and 50 (blue), 5.0 methanol blending ratio, 5.0 MPa, $\phi = 0.5$ and the ammonia conversion rate of 50 %.

increases gradually as the methanol content increases. As to N_2O , it is noteworthy that N_2O changes non-monotonical with methanol ratio, first increasing and then decreasing with the addition of methanol. The peak N_2O concentration seems to locate between SR = 25 % and SR = 50 %. The chemical kinetic and diluted effect are discussed to obtain the deep insight of these trends.

Fig. 17 depicts the key pathways of NO formation and NO reduction at SR = 10 %, (black), 25 % (red) and 50 % (blue) at the ammonia conversion rate of 50 %. As shown in Fig. 17, the flux through route $NH_3 \rightarrow NH_2 \rightarrow H_2NO \rightarrow HNO \rightarrow NO$ decreases from 36.2 % to 30.9 % in proportion when the amount of methanol addition increases from 10 % (black value) to 50 % (blue value). Meanwhile, the NO reduction pathways ($NH_2 + NO = N_2 + H_2O$, $NH_2 + NO = NNH$) are uncompetitive, with the total flux decreasing from 25.3 % to 13.5 %. Given the diluted effect of methanol addition, the branching ratio could not fully explain the mechanism of NO enhancement, as the chain branching ratio is a relative characteristic parameter.

Fig. 18 shows the main reactions in the ROP of NO and N_2O for 10 % (solid lines) and 50 % (dashed lines) methanol blending ratios at the ammonia conversion rate of 50 % ($\omega(NH_3) = 50\%$). It includes generation and consumption reactions of NO and N_2O . As illustrated in Fig. 18

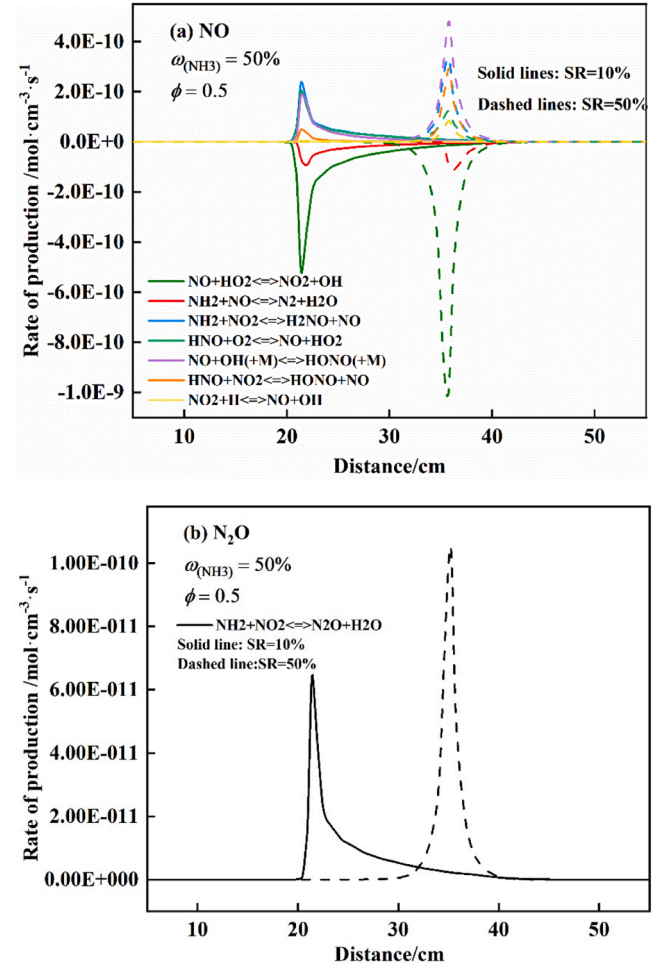


Fig. 18. Main reactions in the ROP of NO and N_2O for 10 % (solid lines) and 50 % (dashed lines) methanol blending ratios at 5.0 MPa, $\phi = 0.5$ and the ammonia conversion rate of 50 %.

(a), the most predominant NO generation reaction is $NH_2 + NO_2 = H_2NO + NO$, followed by $HONO (+M) = NO + OH (+M)$ and $HNO + O_2 = HONO + NO$. NO reduction is dominated by $NO + HO_2 = NO_2 + OH$ and $NH_2 + NO = N_2 + H_2O$. It is obviously all the channels in NO formation are enhanced by methanol addition. The most favorable NO generation

reaction shifts to $\text{HONO} (+\text{M}) = \text{NO} + \text{OH} (+\text{M})$. The more generations of H atom make reaction $\text{NO}_2 + \text{H} = \text{NO} + \text{OH}$ competitive which used to be neglectable in $\text{SR} = 10\%$. Moreover, reaction $\text{NO} + \text{HO}_2 = \text{NO}_2 + \text{OH}$ is enormously accelerated with the increase of methanol blending ratio. It's important to notice that NO_2 is a chain carrier and would be recycled back to NO through reacting with NH_2 . Although it has the largest peak rate of NO consumption, reaction $\text{NO} + \text{HO}_2 = \text{NO}_2 + \text{OH}$ does not affect the mole fraction of NO. Combining the yields of all reactions related to NO, the overall production rate increases along with methanol blending ratio, which is consistent with the results displayed in Fig. 16.

The production rate of N_2O is obviously strengthened by methanol addition from chemistry enhanced as shown in Fig. 17(b). The N_2O formation pathway is favored by $\text{NH}_2 + \text{NO}_2 = \text{N}_2\text{O} + \text{H}_2\text{O}$ and $\text{NH} + \text{NO} = \text{N}_2\text{O} + \text{H}$. Under current conditions, the ROP results show that more than 99.9 % of N_2O is generated through reaction $\text{NH}_2 + \text{NO}_2 = \text{N}_2\text{O} + \text{H}_2\text{O}$, not $\text{NH} + \text{NO} = \text{N}_2\text{O} + \text{H}$. The measured N_2O formation continues to decrease with the addition of methanol. Considering only the chemical effect cannot explain the non-monotonic N_2O formation. In the present experiments, thermal effect is neglected and the reactor can be approximately considered as an isothermal system. An increase in the proportion of blended methanol leads to a corresponding decrease in the proportion of elemental N in the fuel blend, and there is a dilution effect of methanol blending on N_2O production. The dilution effect of methanol also plays a dominant role. As to pure methanol oxidation, NO_x formation through thermal pathways is neglected due to relatively low temperature and no fuel- NO_x is generated. With the reduced N content, the fuel- NO_x that can be generated from a blend of methanol and ammonia becomes less. Therefore, blending more methanol may reduce NO_x formation owing to the dilution effects.

In conclusion, both chemical and dilution effects of methanol blending contribute to NO and N_2O formation. As to NO, the chemical enhancement by methanol addition plays a dominant role in NO formation while dilution effect seems unimportant. Thus, the NO concentration increases monotonously with the increase of methanol blending ratio. However, N_2O yield in the case of $\text{SR} > 25\%$ decreases with increasing methanol content, indicating that the dilution effect is more important than chemical enhancement.

5. Conclusions

In present work, the mole fractions of key intermediates and products during ammonia-methanol mixture were measured at 5.0 MPa, in the temperature range 650–1250 K with different methanol blending ratios. A comprehensive detailed chemical kinetic model for ammonia and methanol was proposed with the revision on the interaction mechanism between C- and N-containing species.

The experimental data showed that NO and N_2O concentrations change non-monotonically with temperature, especially under higher methanol ratio conditions. NO first increases then decreases, and finally increases with temperature. The present model can adequately predict the species concentration in present work, and laminar flame speed, ignition delay time in literature.

With more methanol addition, the ammonia's onset oxidation temperature decreases from 1000 K to around 750 K. The sensitivity analysis shows that pure ammonia oxidation is closely related to NH_3 and NH_2 involved reactions. With methanol addition, reduction in the sensitivity coefficients of the above reactions can be observed. The OH production is related to $\text{CH}_3\text{OH} + \text{HO}_2 = \text{H}_2\text{O}_2 + \text{CH}_2\text{OH}$. In addition, H atom abstraction in CH_3OH by NH_2 facilitates the ammonia oxidation at $\text{SR} = 1\%$, but inhibits the overall reactivity as it converts NH_2 to ammonia.

At 750 K, little ammonia is oxidized, with temperature increasing to 800 K, more ammonia is consumed and generates a large amount of H_2NO radicals. At 850 K, the NO reduction pathways become more dominant. Therefore, NO concentration increases firstly, and then decreases. At much higher temperature, NO concentration is rising again

with more OH and H radicals' production. The most dominant NO generation reaction is $\text{NH}_2 + \text{NO}_2 = \text{H}_2\text{NO} + \text{NO}$, followed by $\text{HONO} (+\text{M}) = \text{NO} + \text{OH} (+\text{M})$ and $\text{HNO} + \text{O}_2 = \text{HONO} + \text{NO}$. All these channels are enhanced by methanol addition.

CRediT authorship contribution statement

Geyuan Yin: Writing – original draft, Validation, Project administration, Methodology, Investigation. **Shujie Shen:** Resources, Methodology, Investigation, Data curation. **Haochen Zhan:** Methodology, Investigation, Data curation. **Erjiang Hu:** Writing – review & editing, Supervision, Methodology. **Hao Zhang:** Writing – review & editing. **Yangyang Bao:** Writing – review & editing. **Zuohua Huang:** Writing – review & editing, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Novelty and significance

This paper investigated the ammonia-methanol mixture oxidation under low-to-intermediate-temperature and high-pressure conditions. It presents the first set of data available for the mole fraction profiles of key intermediates and products at 5.0 MPa, in the temperature range 650–1250 K. In addition, the interaction mechanism between C- and N-containing species was modified to develop an accurate kinetic. The kinetic inside the ammonia-methanol mixture oxidation and NO_x formation was intensive explored.

Acknowledgments

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.combustflame.2024.113654](https://doi.org/10.1016/j.combustflame.2024.113654).

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